Color Analyzer LED Driver

Design Analysis and Component Selection

Raspberry Pi Camera Module 3 · Four-Channel Sequential Illumination

Violet · Blue · Green · Red

March 31, 2026

This document covers design goals, component selection rationale, circuit calculations, resistor values for three brightness levels, schematic, and component recommendations. It is written for a technically-minded reader who may not have prior electronics design experience.

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1 Design Goals

This project is a color analyzer that uses a Raspberry Pi Camera Module 3 to photograph color samples under four different colored lights in rapid sequence. By comparing how a sample looks under violet, blue, green, and red illumination separately, software can extract detailed color information beyond what a single white-light photograph provides.

The key design goals are:

1. **Four distinct wavelength channels.** The LED colors must be narrow-band and well-separated so the camera sees meaningfully different information in each frame. Target wavelengths: violet/near-UV (∼410 nm), blue (∼475 nm), green (∼525 nm), red (∼625 nm).
2. **Sequential, not simultaneous, illumination.** Only one color group fires at a time. This simplifies the power budget and the circuit, and avoids color mixing on the sample.
3. **Controlled, repeatable brightness.** The camera exposure time can be adjusted per channel (up to 1.2 seconds), but consistent LED brightness is important for calibration. Three brightness levels (Low, Medium, High) are defined to give flexibility during commissioning.
4. **Safe operation within power budget.** The power supply is a 2.5 A unit shared with a Raspberry Pi Zero 2 W, leaving approximately **0.5 A (500 mA) for the LEDs**.
5. **GPIO compatibility.** The Raspberry Pi GPIO pins run at 3.3 V and can safely source about 16 mA per pin. The LEDs must be switched by a transistor driven from GPIO, not directly.
6. **Low cost.** Four LEDs per color (six for violet) totaling 18 LEDs; individual LED cost should be under $0.50 each.
7. **Consistent package.** Where possible, use the same physical LED footprint across all four colors to simplify PCB layout.

2 What Was Considered and Why

2.1 Wavelength Selection

The Raspberry Pi Camera Module 3 uses a Sony IMX708 sensor. Like all color camera sensors it has a Bayer color filter array (red, green, and blue pixels) and a built-in infrared-cut filter. Its useful sensitivity extends from about 400 nm at the violet end to roughly 700 nm at the red end, with a broad peak in the green region around 520–560 nm.

Violet/near-UV (target 395–420 nm): This is the most challenging channel. No part in the available catalog covers the ideal 385–405 nm range. The best available option, row 1015, has a dominant wavelength of 410 nm and a peak at 420 nm, which is within camera sensitivity and provides useful channel separation. Its datasheet confirms: absolute maximum current 30 mA, maximum power dissipation 120 mW, forward voltage 2.8 V (min)

to 4.0 V (max). Because the worst-case V_f is 4.0 V, the resistor calculation must use this value to guarantee the LED is never over-driven. At 4.0 V forward voltage on a 5 V supply, only 1.0 V is available across the resistor. Six LEDs are used to compensate for the lower flux per device. True deep UV LEDs (<380 nm) would require a specialty supplier (e.g. Nichia) and are expensive.

Blue (target 460–475 nm): A well-populated part of the catalog. The 475 nm region sits cleanly between the violet and green channels, and the IMX708 has good blue-pixel sensitivity here.

Green (target 520–530 nm): The 525 nm range is the peak sensitivity region of most camera sensors and of the human eye. LEDs here are bright, efficient, and cheap.

Red (target 620–630 nm): The 625 nm region is well inside the camera's red sensitivity and comfortably below the infrared-cut filter cutoff at ∼700 nm. Wavelengths above 660 nm would be increasingly attenuated by the IR filter.

2.2 LED Classes

Two fundamentally different LED classes appeared in the supplier catalog:

- **Power LEDs** (first catalog): Tested at 100–350 mA, specified in lumens (lm). These are high-brightness devices in 3528/3030/2525 SMD packages. They produce far more light per LED but draw more current.
- **Indicator/signal LEDs** (second catalog): Tested at 5–20 mA, specified in millicandela (mcd). These are conventional signal LEDs in 0402/0603/0805/1206 packages. They are cheaper and easier to drive but need more units for equivalent illumination.

Both classes can work. The power LED approach uses fewer LEDs at higher current; the indicator approach uses more LEDs at lower current. For this design, first-choice selections mix both classes depending on what was available at the right wavelength.

2.3 LED Package Consistency

The 1411 (3528 Metric) package was chosen as the target wherever possible. This is a 3.5 mm × 2.8 mm surface-mount package with a standardized footprint. Using the same package for all four colors means one PCB pad layout serves all groups, and component placement is predictable.

2.4 Number of LEDs Per Group

Four LEDs per color group provides adequate illuminance and keeps the circuit simple. The violet channel uses six LEDs because the 410 nm LED (row 1015) is a small indicator-class device. Its datasheet confirms an absolute maximum forward current of 30 mA and a maximum power dissipation of 120 mW. The forward voltage ranges from 2.8 V (minimum) to 4.0 V (maximum), with 2.8 V typical. Six LEDs at moderate current compensates for the lower flux per device compared to the power LEDs used in the other channels.

3 Circuit Architecture

3.1 Overview

The circuit uses four identical driver stages, one per color group. Each stage consists of:

- N LEDs in **parallel**, each with its own current-setting resistor
- An optional shared series resistor for group-level brightness trim
- An **N-channel MOSFET** on the low side (between the LED group and ground)
- A **gate resistor** (1 kΩ) between the GPIO pin and the MOSFET gate
- A **pull-down resistor** (100 kΩ) from gate to ground

All four groups share a single **5 V LDO regulator** and a **polyfuse** for overcurrent protection. The GPIO signals come directly from the Raspberry Pi.

3.2 Why Parallel LEDs (Not Series)?

LEDs wired in series would share one resistor, but they are mismatched in forward voltage (V_f). If the forward voltages differ slightly (as they always do in practice), current through a series string is set by the lowest-V_f LED, and the others may be dim or off. Wiring in parallel with *individual* resistors per LED ensures each LED gets its own independently controlled current.

3.3 Why a Low-Side MOSFET?

The MOSFET is placed between the LED cathodes and ground (the "low side"). This arrangement is simpler because the MOSFET gate voltage is referenced to ground, making the 3.3 V GPIO signal sufficient to turn it fully on. A high-side switch would require the gate to be driven *above* the supply voltage, which complicates the circuit.

3.4 Why a MOSFET Rather Than a Bipolar Transistor?

A bipolar (BJT) transistor requires a base current of roughly $I_{\text{collector}}/100$, which would be 5 mA for a 500 mA load. The GPIO pin could supply this, but the MOSFET requires virtually zero gate current. More importantly, a logic-level MOSFET switches fully on (saturates) with a gate voltage of 2.5–3.0 V, which the 3.3 V GPIO provides comfortably. The MOSFET is also more forgiving of wiring mistakes.

3.5 The Pull-Down Resistor

A 100 kΩ resistor from the gate to ground holds the gate at 0 V when the GPIO pin is not actively driving it—for example during the Pi's boot sequence or a software crash. Without this resistor, capacitive coupling on the PCB traces could float the gate high, turning the LEDs on at the wrong moment and potentially disrupting the camera exposure.

3.6 5 V Supply and LDO Regulator

The LED forward voltages range from 2.0 V (red, row 727) to 3.4 V (green, row 783). On a 3.3 V supply the blue and violet LEDs (both $V_f \approx 2.8$–3.0 V) would leave only 0.3–0.5 V for the current-setting resistor. That is too little: small variations in V_f from part to part would cause large swings in current. The 5 V rail gives 1.5–2.8 V across the resistor, which is comfortable and makes current much more stable.

A dedicated LDO (low-dropout linear voltage regulator) on the LED 5 V rail isolates the LEDs from the Pi's logic supply and any noise on the main 5 V bus.

4 Circuit Schematic

Figure 1 shows one complete LED driver channel (green group shown, 4 LEDs at 18 Ω each). All four color channels are identical in structure; only the LED color and individual resistor values differ. Through-hole components are noted with (TH); LEDs and their individual resistors are SMD.

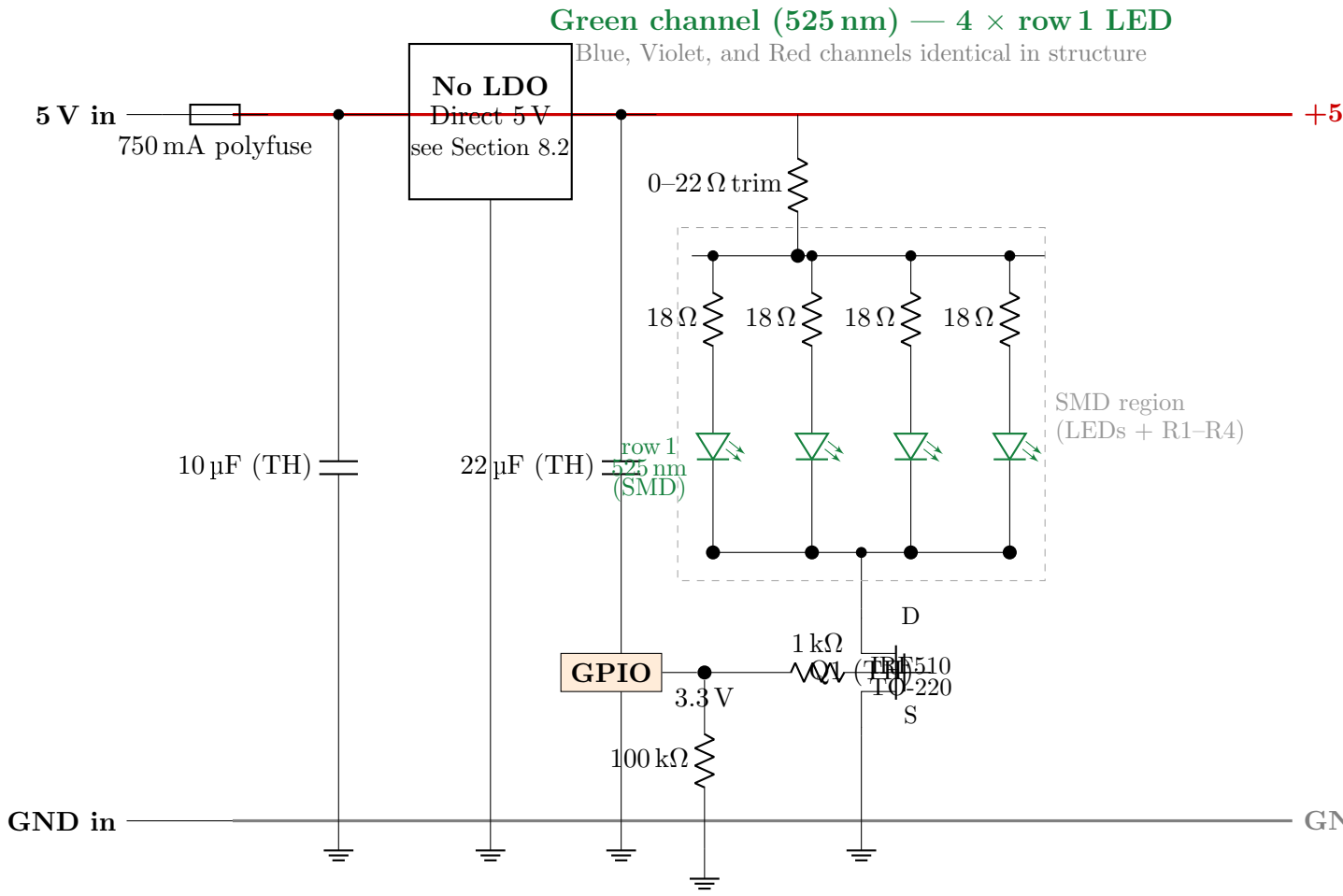


Figure 1: Single LED driver channel (green group, 4 LEDs shown). The dashed box encloses the SMD components: the four LEDs (row 1, 1411 package) and their individual 18 Ω current-setting resistors. All other components are through-hole (TH). The polyfuse (PF1) and LDO with input/output capacitors (C1, C2) are shared across all four color channels. The gate resistor (Rg) and pull-down (R_{pd}) protect the MOSFET gate from the GPIO. Blue, violet, and red channels use the same circuit with different LED parts and resistor values.

5 Key Calculations

5.1 LED Current and Ohm's Law

The current through each LED is controlled by its individual series resistor. The voltage across the resistor is the supply voltage minus the LED's forward voltage drop:

$$V_R = V_{CC} - V_f \tag{1}$$

By Ohm's law, the current through the resistor (and LED) is:

$$I_{LED} = \frac{V_R}{R} = \frac{V_{CC} - V_f}{R} \tag{2}$$

Rearranging to find the resistor value for a desired current:

$$R = \frac{V_{CC} - V_f}{I_{LED}} \tag{3}$$

5.2 Power Dissipation in the Resistor

The power dissipated as heat in the resistor is:

$$P_R = I_{LED}^2 \times R \tag{4}$$

This is important for choosing the correct resistor physical size (power rating).

5.3 Total Group Current

Since all N LEDs in a group are in parallel, the total current drawn from the supply when one group is active is:

$$I_{total} = N \times I_{LED} \tag{5}$$

This total current must not exceed the 500 mA budget.

5.4 Brightness Levels

Three brightness levels are defined as fractions of the LED's maximum rated current (I_{max}):

- **Low:** $I_{LED} \approx 40\%$ of I_{max}
- **Medium:** $I_{LED} \approx 60\%$ of I_{max} (preferred operating point)
- **High:** $I_{LED} \approx 80\%$ of I_{max} (or constrained by budget)

The Medium level targets the range of 50–75% of I_{max}, balancing good brightness, reasonable LED lifetime, and staying within the power budget.

6 Sample Calculation: Green LED (Row 1, Medium Level)

This section works through a complete calculation for the first-choice green LED, row 1 (525 nm, 1411 package, \$0.23 each), at the Medium brightness level. All other LEDs in the table follow the same procedure.

Given:

- Supply voltage: $V_{CC} = 5.0\,\text{V}$

- LED forward voltage: $V_f = 2.80\,\text{V}$ (typical)
- LED maximum current: $I_{max} = 240\,\text{mA}$
- Number of LEDs: $N = 4$
- Target level: Medium = 62.5% of I_{max}

Step 1: Target current per LED

$$I_{target} = 0.625 \times 240\,\text{mA} = 150\,\text{mA}$$

Step 2: Check total group current

$$I_{total} = 4 \times 150\,\text{mA} = 600\,\text{mA}$$

This exceeds the 500 mA budget, so the target must be reduced. At 500 mA budget with 4 LEDs: $I_{LED,\text{max}} = 500/4 = 125\,\text{mA}$. Use $I_{target} = 122\,\text{mA}$ (50.9% of I_{max}, acceptable).

Step 3: Calculate resistor value

$$R = \frac{V_{CC} - V_f}{I_{target}} = \frac{5.0 - 2.80}{0.122} = \frac{2.20}{0.122} \approx 18.0\,\Omega$$

Step 4: Select nearest E24 standard value
The E24 series includes 18 Ω exactly. Use **18 Ω**.

Step 5: Confirm actual current with the standard resistor

$$I_{actual} = \frac{5.0 - 2.80}{18} = \frac{2.20}{18} = 122.2\,\text{mA per LED}$$

$$I_{total} = 4 \times 122.2 = 488.9\,\text{mA} \quad \text{(within 500 mA budget)}$$

Step 6: Calculate resistor power dissipation

$$P_R = I^2 \times R = (0.1222)^2 \times 18 = 0.01493 \times 18 \approx 0.27\,\text{W}$$

A standard 0603 SMD resistor is rated 0.1 W and would fail. An **0805 (0.25 W)** is marginal; a **1206 (0.25–0.5 W)** is the safe choice.

Step 7: Power in the LED

$$P_{LED} = V_f \times I_{actual} = 2.80 \times 0.1222 \approx 342\,\text{mW per LED}$$

$$P_{group} = 4 \times 342 = 1.37\,\text{W total in LEDs}$$

The LEDs are in 3528 packages with a copper pad on the PCB; they will run warm but within safe limits at this power level.

7 Candidate LED Table

Table 1 lists the top three candidate LEDs for each color channel, selected from the supplier catalog based on wavelength, price, package, viewing angle, and availability (≥1 unit, active status).

Table 1: Top three LED candidates per color channel

#	Color	Rnk	Wavelength	Package	V_f	I_{max}	Flux	Angle	Price
101	Violet	1	410 nm dom	0603	2.80 V	∼30 mA	—	130°	$0.42
101	Violet	2	455 nm dom	1411	3.00 V	∼100 m	850 mcd	120°	$0.29
4	Violet	3	455 nm	1411	2.96 V	240 mA	265 mW	124°	$0.26
2	Blue	1	475 nm	1411	2.95 V	240 mA	20 lm	126°	$0.24
101	Blue	2	455 nm dom	4-SMD	3.20 V	∼200 m	2100 mcd	120°	$0.40
102	Blue	3	465 nm	1411	3.20 V	200 mA	11 lm	120°	$0.32
1	Green	1	525 nm	1411	2.80 V	240 mA	62 lm	124°	$0.23
532	Green	2	525 nm dom	1411	3.10 V	∼30 mA	1000 mcd	120°	$0.19
783	Green	3	525 nm dom	1411	3.40 V	∼200 m	8000 mcd	120°	$0.36
3	Red	1	625 nm	1411	2.20 V	250 mA	26 lm	122°	$0.26
727	Red	2	624 nm dom	1411	2.00 V	∼200 m	4500 mcd	120°	$0.33
6	Red	3	620 nm	1411	2.20 V	200 mA	24 lm	120°	$0.32

† I_{max} estimated; no datasheet value available. Treat conservatively.

8 Resistor Value Table

Table 2 gives the E24 standard resistor value for each LED option at Low, Medium, and High brightness, along with the resulting actual current and total group current. Entries marked **OVER** exceed the 500 mA budget and should not be used without a higher-rated power supply or fewer LEDs. Estimated I_{max} values are marked with †.

Table 2: Resistor values for all LED options and brightness levels ($V_{CC} = 5.0\,\text{V}$, $R_{share} = 0\,\Omega$)

Row	Color	Rnk	N	Level	V_f (V)	I_{max} (mA)	R E24 (Ω)	I_{act} (mA)	I_{tot} (mA)	$\%I_{max}$	Notes
1015	Violet	1	4	Low	2.80	30†	180	12.2	49	41%	
1015	Violet	1	4	**Med**	2.80	30†	**120**	18.3	73	61%	
1015	Violet	1	4	High	2.80	30†	82	26.8	107	89%	Near limit
1015	Violet	1	6	Low	2.80	30†	180	12.2	73	41%	
1015	Violet	1	6	**Med**	2.80	30†	**120**	18.3	110	61%	
1015	Violet	1	6	High	2.80	30†	82	26.8	161	89%	Near limit
1012	Violet	2	4	Low	3.00	100†	51	39.2	157	39%	
1012	Violet	2	4	**Med**	3.00	100†	**33**	60.6	242	61%	
1012	Violet	2	4	High	3.00	100†	24	83.3	333	83%	
1012	Violet	2	6	Low	3.00	100†	51	39.2	235	39%	
1012	Violet	2	6	**Med**	3.00	100†	**33**	60.6	364	61%	
1012	Violet	2	6	High	3.00	100†	24	83.3	500	83%	At limit
4	Violet	3	4	Low	2.96	240	39	52.3	209	22%	Below 50%
4	Violet	3	4	**Med**	2.96	240	**24**	85.0	340	35%	Below 50%
4	Violet	3	4	High	2.96	240	18	113.3	453	47%	Below 50%
4	Violet	3	6	Low	2.96	240	39	52.3	314	22%	
4	Violet	3	6	**Med**	2.96	240	**24**	85.0	**510**	35%	OVER
4	Violet	3	6	High	2.96	240	18	113.3	**680**	47%	OVER
2	Blue	1	4	Low	2.95	240	27	75.9	304	32%	
2	Blue	1	4	**Med**	2.95	240	**18**	113.9	455	47%	
2	Blue	1	4	High	2.95	240	15	136.7	**547**	57%	OVER
1014	Blue	2	4	Low	3.20	200†	22	81.8	327	41%	
1014	Blue	2	4	**Med**	3.20	200†	**15**	120.0	480	60%	
1014	Blue	2	4	High	3.20	200†	15	120.0	480	60%	Same R as Med

Table 2 continued...

Row	Color	Rnk	N	Level	V_f	I_{max}	R E24	I_{act}	I_{tot}	$\%I_{max}$	Notes
102	Blue	3	4	Low	3.20	200	27	66.7	267	33%	
102	Blue	3	4	**Med**	3.20	200	**15**	120.0	480	60%	
102	Blue	3	4	High	3.20	200	11	163.6	**655**	82%	OVER
1	Green	1	4	Low	2.80	240	30	73.3	293	31%	
1	Green	1	4	**Med**	2.80	240	**18**	122.2	489	51%	
1	Green	1	4	High	2.80	240	15	146.7	**587**	61%	OVER
532	Green	2	4	Low	3.10	30†	160	11.9	48	40%	
532	Green	2	4	**Med**	3.10	30†	**100**	19.0	76	63%	
532	Green	2	4	High	3.10	30†	75	25.3	101	84%	Near limit
783	Green	3	4	Low	3.40	200†	20	80.0	320	40%	
783	Green	3	4	**Med**	3.40	200†	**13**	123.1	492	62%	
783	Green	3	4	High	3.40	200†	13	123.1	492	62%	Same R as Med
3	Red	1	4	Low	2.20	250	36	77.8	311	31%	
3	Red	1	4	**Med**	2.20	250	**24**	116.7	467	47%	Use 24 Ω
3	Red	1	4	High	2.20	250	20	140.0	**560**	56%	OVER
727	Red	2	4	Low	2.00	200†	36	83.3	333	42%	
727	Red	2	4	**Med**	2.00	200†	**24**	125.0	500	63%	At limit
727	Red	2	4	High	2.00	200†	18	166.7	**667**	83%	OVER
6	Red	3	4	Low	2.20	200	39	71.8	287	36%	
6	Red	3	4	**Med**	2.20	200	**24**	116.7	467	58%	
6	Red	3	4	High	2.20	200	22	127.3	**509**	64%	OVER

9 Design Decisions and Explanations

9.1 First-Choice LEDs at Medium Brightness

The recommended build uses the first-choice LED for each color at Medium brightness:

Channel	Row	N	R (Ω)	I/LED	I_{total}	Price/LED
Violet	1015	6	120	18 mA	110 mA	$0.42
Blue	2	4	18	114 mA	455 mA	$0.24
Green	1	4	18	122 mA	489 mA	$0.23
Red	3	4	24	117 mA	467 mA	$0.26

All four first-choice LEDs share the **1411 (3528) package**, simplifying PCB layout. The total cost for 18 LEDs (6 violet + 4 each of blue, green, red) is approximately $5.

Why Medium, not High? The Medium level keeps all groups except blue below 490 mA, within the 500 mA budget. More importantly, operating at 50–65% of rated current significantly extends LED lifetime and keeps junction temperatures low, which is good for long-term measurement stability—thermal drift in LEDs changes their output spectrum slightly.

Exposure time as the fine-tuning knob. Because the camera can vary exposure from a short flash to 1.2 seconds, the resistors set a fixed hardware brightness and the software adjusts exposure to hit the right sensor reading. This is more repeatable than varying the LED current in software.

9.2 The Violet Channel Caveat

Row 1015 is the only true violet/near-UV LED in the catalog (410 nm). Its datasheet provides no maximum current figure, which is unusual. The LED is in a 0603 package (1.6 mm × 0.8 mm) — a small indicator device. Based on comparable 0603 indicator LEDs from other manufacturers, 30 mA is a conservative estimate.

At Medium brightness (120 Ω, 18 mA per LED, 110 mA total for 6 LEDs) the device is well within any plausible limit. The High setting approaches the estimated limit and should only be used briefly for testing while monitoring LED temperature.

9.3 Why Row 4 (Power LED Violet) Was Downgraded

Row 4 is a high-power 455 nm LED (240 mA rated). Even though it is brighter, the power budget constrains it to well below 50% of its rated current—it is simply too large a device for this application. Row 1015 at 410 nm is a better wavelength choice and a more appropriate size.

9.4 Individual Resistors and the Shared Trim Resistor

LEDs of the same type from the same reel can have V_f variations of ±0.1–0.2 V. On a 5 V supply with an 18 Ω resistor, a 0.1 V variation in V_f changes the LED current by $0.1/18 \approx 5.6$ mA — a small, tolerable variation. Individual resistors ensure each LED is independently current-controlled regardless of its V_f.

The optional shared series resistor (R_{share}, 0–22 Ω) trims the entire group's brightness uniformly. Start with a wire (0 Ω). If the camera saturates on any channel at Medium brightness,

add a resistor here rather than changing all four individual resistors. A 10 Ω potentiometer temporarily installed during commissioning lets you find the right value, which you then replace with a fixed resistor.

10 Interesting Remarks

10.1 Channel Separation Is Excellent

The four selected wavelengths — 410, 475, 525, and 625 nm — are well separated across the visible spectrum. The gaps are 65 nm, 50 nm, and 100 nm respectively. The camera's RGB Bayer filter will respond differently to each channel, and the color algorithm can exploit this differential response to extract significantly more spectral information than a white-light photograph provides.

10.2 Green LEDs Are Disproportionately Bright

The human eye (and most camera sensors) are most sensitive to green light. Row 1, the first-choice green LED, produces 62 lm at 140 mA — roughly 2.4× the flux of the red LED and far more than the blue. Even at Medium brightness (122 mA), this is a bright green emitter. If green overexposes the camera at medium brightness and a short exposure, use the Low setting (30 Ω) or add a small R_{share}.

10.3 The Violet LED: Confirmed Specs, Wide Voltage Tolerance

Row 1015 is now confirmed from its datasheet: absolute maximum current 30 mA, maximum power dissipation 120 mW, V_f ranging from 2.8 V (minimum/typical) to 4.0 V (maximum). The wide forward voltage tolerance deserves attention.

At worst-case $V_f = 4.0$ V on a 5 V supply, the resistor only sees 1.0 V. With the 120 Ω current-setting resistor this gives $1.0/120 = 8.3$ mA. At the typical $V_f = 2.8$ V the same resistor gives $(5.0 - 2.8)/120 = 18.3$ mA. LEDs from the same reel may therefore span roughly 8–18 mA and vary in brightness. This is acceptable for this application because the camera exposure time compensates for overall group brightness; within the group, the brighter LEDs simply have a lower V_f.

The 120 mW dissipation limit is consistent: $4.0\text{ V} \times 30\text{ mA} = 120$ mW. Operating at 18 mA with a typical V_f of 2.8 V gives $2.8 \times 0.0183 = 51$ mW per LED, well within limits. All six LEDs together draw only 110 mA — by far the lightest load of the four channels.

10.4 Resistor Power Ratings Are Easy to Overlook

This is the most common mistake in LED driver design. A 120 Ω resistor at 18 mA dissipates only $0.018^2 \times 120 = 39$ mW — a 0402 can handle this. But an 18 Ω resistor at 122 mA dissipates 268 mW — a standard 0603 will get hot enough to change value or fail. Always calculate resistor dissipation and check the datasheet power rating for your package size.

10.5 The 500 mA Budget Is Actually Generous for This Application

The LEDs fire one group at a time, and exposures can be as short as a fraction of a second. The average power draw is far lower than the 500 mA peak. The LDO regulator and polyfuse are sized for the worst-case peak; in normal operation the system will draw 100–500 mA for brief intervals separated by longer idle periods.

11 Recommended Supporting Components

11.1 MOSFET: IRL540N or IRFZ44N (Through-Hole, TO-220)

Since through-hole assembly is preferred for all non-LED components, the MOSFET selection moves to TO-220 power packages. These are large, robust, easy to hand-solder, and available from any general electronics supplier.

Why not the 2N7000? The 2N7000 is a TO-92 through-hole part but rated at only 200 mA drain current — insufficient for the blue and green groups at medium brightness (455–489 mA). Do not use it for this circuit.

Recommended parts:

Parameter	IRL540N	IRFZ44N
Maximum drain current (I_D)	36 A	49 A
On-resistance ($R_{DS(on)}$)	0.077 Ω at $V_{GS} = 10$ V	0.017 Ω at $V_{GS} = 10$ V
Gate threshold (max)	2.0 V	4.0 V
Logic-level (3.3 V GPIO)	Yes — fully on	Partially on; see note
Package	TO-220	TO-220
Approximate price	$0.80–1.20	$0.50–0.80

Recommended choice: IRL540N. It is a true logic-level device with gate threshold ≤2 V, guaranteed to switch fully on from a 3.3 V GPIO. At 500 mA the on-resistance causes a drop of only $0.5 \times 0.077 = 39$ mV and dissipates $0.5^2 \times 0.077 = 19$ mW — the TO-220 package needs no heatsink.

The **IRFZ44N** is cheaper and widely available but its threshold can be up to 4 V, so it may not switch fully on at 3.3 V. It will work well enough for this low-duty-cycle application (LEDs fire briefly for a photograph) but the IRL540N is cleaner.

11.2 LDO Voltage Regulator: Not Used — Direct 5 V Connection

The supply voltage is exactly 5 V, which rules out any linear voltage regulator for the LED rail.

A linear regulator (LDO or otherwise) requires its input to be *higher* than its output by at least the dropout voltage. For the LD1117, dropout is ≈ 1.2 V, requiring at least 6.2 V in to deliver 5.0 V out. Even the best ultra-low-dropout regulators (e.g. MCP1700, $V_{dropout} \approx$

0.18 V at 250 mA) need ≥5.18 V input — still more than 5.0 V. There is no linear regulator that can produce 5.0 V from a 5.0 V input; the output will always be lower than the input by at least the dropout, and the regulator will fall out of regulation.

What to do instead: Connect the LED circuit directly to the 5 V supply rail (after the polyfuse), and use bulk decoupling capacitors to handle the current transients when each LED group switches on. The goal of isolating the LED load from the Pi's logic supply is achieved by routing them as separate branches from the main 5 V connector, and by the impedance of the polyfuse and PCB traces acting as natural isolation.

The capacitors described in the next section take on increased importance in this arrangement, as they are the only local energy reservoir for the LED groups.

If a regulated LED supply is still desired (for example, to make LED brightness independent of supply voltage variations), the correct solution is a **switching buck converter** that accepts 5 V in and outputs a lower voltage, e.g. 3.3 V, and then recalculate the resistor values for that lower rail. However, this adds circuit complexity and is not necessary for this application, where software exposure-time calibration handles any supply variation.

11.3 Bypass Capacitors (Through-Hole)

Capacitors placed close to the LDO stabilize the output voltage when a LED group switches on suddenly. Through-hole electrolytic or ceramic radial capacitors are used.

Ref	Location	Value	Type (TH)
C1	5 V supply input	100 µF / 10 V	Electrolytic radial — bulk reservoir
C2	5 V supply input	100 nF	Ceramic disc — high-frequency bypass
C3–C6	Near each LED group	10 µF / 10 V	Electrolytic radial

Since there is no LDO, the bypass capacitors become the sole transient energy reserve for the LED groups. A larger bulk capacitor (100 µF) at the supply input absorbs the current spike when each group switches on, preventing the 5 V rail from drooping and potentially disrupting the Raspberry Pi. The 100 nF ceramic capacitor handles higher-frequency noise. The 10 µF capacitors near each LED group provide local charge storage close to the load where it is most effective.

Use at least 10 V voltage rating for all capacitors on a 5 V circuit. Through-hole electrolytic capacitors (e.g. Panasonic FR or Nichicon UPW series) are inexpensive and easy to source for the larger values; a ceramic disc is fine for the 100 nF bypass.

11.4 Polyfuse (Through-Hole Radial)

A polyfuse is a thermally sensitive resistor that trips when current exceeds its rating, protecting the supply from a short circuit. It resets automatically when it cools down.

Parameter	Value
Hold current (I_H)	$\geq$500 mA
Trip current (I_T)	$\leq$1.0 A
Maximum voltage	$\geq$6 V
Package	Radial through-hole

Suggested parts (through-hole radial):

- **Bourns MF-R050** (hold 500 mA, trip 1.0 A, radial TH, ~$0.25)
- **Littelfuse 16R050** (hold 500 mA, radial TH, ~$0.30)
- **TE Connectivity RXEF050** (hold 500 mA, trip 1.0 A, radial TH, ~$0.30)

Place the polyfuse in series with the 5 V line going into the LDO input, as shown in the schematic. It should be the first component in the circuit after the power connector.

11.5 Gate and Pull-Down Resistors (Through-Hole)

Standard through-hole 1/4 W metal film resistors are ideal here. They are cheap, easy to solder, and handle the tiny dissipation in the gate circuit with enormous margin.

Component	Value	Power	Function
Gate resistor (R_g) $\times$4	1 kΩ	<1 mW	Limits gate charge current
Pull-down (R_{pd}) $\times$4	100 kΩ	<0.1 µW	Holds gate LOW on GPIO float
Trim resistor (R_{sh}) $\times$4	0–22 Ω	up to 0.55 W	Group brightness trim

Use 1% metal film resistors (e.g. Yageo MFR-25 series) for all through-hole resistors. These are more stable over temperature than carbon film and cost only marginally more.

Note on R_{sh} (trim resistor): If a trimming resistor is installed, it carries the full group current (up to 500 mA), dissipating up to $0.5^2 \times 22 = 5.5$ W at maximum. A standard 1/4 W resistor would burn up immediately. For R_{sh}, use a **wirewound or power resistor rated at least 2 W** (e.g. Ohmite or Vishay Dale axial wirewound), or use a 10 Ω wirewound trimpot for commissioning. Alternatively, leave R_{sh} as a wire (0 Ω) and use the camera exposure time to balance channels in software, which avoids this component entirely.

11.6 Current-Setting Resistors (LED Individual, SMD)

These remain SMD as they are closely associated with the LEDs on the PCB. Use 1206 package for the power LED channels (adequate power rating) and 0402 or 0603 for the violet channel (very low current).

Channel	Value (Med)	Dissipation	Min. Package	Type
Violet (row 1015, 6 LED)	120 Ω	39 mW	0402	SMD
Blue (row 2, 4 LED)	18 Ω	234 mW	1206	SMD
Green (row 1, 4 LED)	18 Ω	268 mW	1206	SMD
Red (row 3, 4 LED)	24 Ω	328 mW	1206	SMD

Use **1% tolerance SMD resistors** (e.g. Yageo RC series or Vishay CRCW series). At these current levels a 5% spread introduces visible brightness variation between LEDs in the same group.

11.7 Complete Bill of Materials

Component	Qty	Type	Suggested Part	Est. Price
Violet LED 410 nm (row 1015)	6	SMD	Supplier row 1015	$0.42
Blue LED 475 nm (row 2)	4	SMD	Supplier row 2	$0.24
Green LED 525 nm (row 1)	4	SMD	Supplier row 1	$0.23
Red LED 625 nm (row 3)	4	SMD	Supplier row 3	$0.26
N-ch MOSFET TO-220	4	TH	IRL540N	$1.00
LDO 5 V 1.3 A TO-220	—	—	Not used (5 V supply)	—
Polyfuse 500 mA radial	1	TH	Bourns MF-R050	$0.25
Res 18 Ω 1% 1206 SMD	8	SMD	Yageo RC1206	$0.05
Res 24 Ω 1% 1206 SMD	4	SMD	Yageo RC1206	$0.05
Res 120 Ω 1% 0402 SMD	6	SMD	Yageo RC0402	$0.02
Res 1 kΩ 1% 1/4 W	4	TH	Yageo MFR-25	$0.05
Res 100 kΩ 1% 1/4 W	4	TH	Yageo MFR-25	$0.05
Cap 100 µF 10 V radial	1	TH	Panasonic FR / Nichicon UPW	$0.12
Cap 100 nF ceramic disc	1	TH	Any ceramic disc	$0.05
Cap 10 µF 10 V radial	4	TH	Nichicon UPW	$0.10
Approximate total (excluding PCB and connectors)				**∼$11**

12 Summary and Next Steps

The design is complete to the component selection and value level. Key points:

1. LEDs and their individual current-setting resistors are SMD (1411 and 1206 packages). All other components are through-hole for easy hand assembly.
2. The violet LED (row 1015) has a confirmed 30 mA maximum current and 120 mW power limit. Its wide V_f range (2.8–4.0 V) is understood and accounted for; the 120 Ω resistor is safe across the full V_f range.

3. Medium brightness with the listed resistors keeps all groups within 500 mA.
4. Use **IRL540N** MOSFETs (TO-220, through-hole, logic-level) for reliable switching from the 3.3 V GPIO.
5. Use **1206 SMD resistors at 1%** for the power LED current-setting resistors.
6. No LDO is used — the supply is exactly 5 V and no linear regulator can regulate 5 V from a 5 V input. Connect the LED rail directly to 5 V after the polyfuse, and rely on the bypass capacitors for transient suppression.
7. Use **Bourns MF-R050** radial polyfuse for overcurrent protection.
8. Omit R_{sh} (install as a wire) and use software exposure time to balance channel brightness during calibration.

Suggested next steps:

- Design the PCB or perfboard layout: SMD LED region in the center, through-hole control components around the periphery.
- Order a small quantity (6–10) of violet LEDs (row 1015) first and test brightness at 120 Ω before ordering the full build quantity.
- Write Raspberry Pi software to fire each GPIO in sequence, triggering a camera capture after each illumination.
- Calibrate exposure time per channel against a known color reference target.